Roll No. | | | | | | | |

Mid-term Semester Examination 2025

Name of the Program: M. Tech. (Biomedical Engineering) *Semester: Ist*
Name of the Course: Biomaterials Engineering *Course Code:* ***BEM 101***

Time: 1-1/2 Hour Maximum Marks: 50

Note:-

(i) Answer all the questions by choosing any one of the sub questions.
(ii) Each question carries 10 marks

Q1	(10marks)	**CO1**
(a)	How do physiological fluids impact the characteristics of biomaterials? Provide examples to explain these effects.	
	or	
(b)	What are biomaterials, and what essential criteria must an ideal biomaterial fulfill?	
Q2	(10 marks)	**CO1**
(a)	Describe the different categories of biomaterials and provide appropriate examples for each.	
	or	
(b)	What are the key surface properties of biomaterials, and how do they affect their biocompatibility?	
Q3	(10 marks)	**CO1, CO2**
(a)	How do the mechanical properties of metals, ceramics, and polymers used as biomaterials differ from each other?	
	or	
(b)	Identify the commonly used metallic materials for implants and explain the benefits of each.	
Q4	(10 marks)	**CO1, CO2**
(a)	Describe the mechanical properties that biomaterials need to have for use in load-bearing applications.	
	or	
(b)	Describe the response of host tissue to biometals and explain how corrosion influences this interaction.	
Q5	(10 marks)	**CO2**
(a)	What difficulties are encountered when designing dental implants using metallic biomaterials?	
	or	
(b)	Explain the various types of soft tissue replacement implants and the typical materials used for each.	

Note for the question paper setters:

- **Question paper should cover all the COs of the course.**